POLICY REPORT · REFORM PROPOSITION

The Regenerative Municipality

Friction Economics, Digital Legal Safeguards, and Repairable Municipal Capacity

REFORM PROPOSITION FOR KL — LOCAL GOVERNMENT DENMARK AND THE MUNICIPAL SECTOR

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1. Editorial Note and Claim Status

Version history: The first public version of this report was published in May 2026. Version 1.1 fully revised the report in August 2026. Version 1.1.1 from September 2026 tightened the language without changing the report's proposals. Later revisions are identified in the current version line.

The Danish welfare state does not face a funding problem alone. It also faces a capacity problem. Municipalities may have substantial human, organisational, and financial resources tied up in handovers, rework, unclear chains of responsibility, defensive control routines, opaque digital systems, and repeated compensation for problems that never become sufficiently visible as coherent work fields.

This report advances a reform proposition: that some of this tied-up capacity may be identifiable and reducible without weakening legal safeguards, professional quality, working conditions, citizens' access to support, or the physical and ecological conditions on which municipal welfare depends.

> It is a proposition, not a documented national effect.

The report does not present a validated savings potential and does not claim that municipal financial pressures can be resolved through friction reduction. Nor does it claim that administrative tasks are generally unnecessary, that AI automatically frees capacity, or that a local pilot can be generalised to other municipalities.

Parts of the architecture have empirical, organisational, and methodological precursors. But friction economics, G1/G2/G3, precise democratic sensing, the Orwell test, ecological capacity as a municipal pilot field, and the proposed cross-municipal learning architecture have not been validated as one integrated municipal model.

The purpose of the pilot track is therefore to allow municipal practice to correct the proposition — not to confirm an effect assumed in advance.

What regenerative means in this report

The term regenerative is not used as green branding or as a claim that a municipality can make all its activities analogous to nature. A municipality is regenerative to the extent that its operations, economy, and institutional development protect and rebuild the capacities on which future public-service delivery depends.

These include employees' ability to exercise professional judgement, the citizen's ability to understand and contest, the organisation's ability to detect and correct errors, democracy's ability to locate responsibility, and the physical and ecological capacity of a place to absorb pressure.

Regeneration does not mean that every intervention must produce a positive net effect within 90 days. It means that the municipality examines whether its solutions maintain or erode their own human, institutional, and ecological foundations.

> The regenerative municipality is not primarily the expansive municipality. It is the municipality that can see what its delivery of public tasks consumes, halt a harmful development, and repair the conditions on which future welfare depends.

The report's place in the Green Papers inquiry

The Regenerative Municipality is closely related to several other Green Papers, but it has its own institutional task.

Municipal Work as Nature describes the ecology of the work field: rhythm, load, professional judgement, the employee as organisational buffer, and the practical work around Flow · Friction · Sensitivity. This report, by contrast, describes the cross-municipal architecture for reform, rights, and learning.

The Correction Loop carries the formal discipline of accountability: human responsibility, the last decision impulse, stop, and correction. Penguin Dashboard describes a shared reading surface on which different considerations can be seen without automatically compensating for one another. Knowing From the Ground develops situated, fallible, and correctable knowledge. Eve & Adam, and the Penguins describes the research and green-economics lineage. Regenerative Reciprocity addresses flow, support, and governance architecture, while Moral Biology examines the material and biological conditions of human capacity for responsibility.

These texts are methodological and conceptual precursors. They do not constitute independent empirical validation of this report's own claims.

Editorial practice: AI-assisted and human-responsible. The shared method and location of responsibility are described in Methods / Editorial Practice and AI, Aligned Intelligence & Human Responsibility.

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2. Executive Brief: The First Yes

Municipalities operate within a persistent field of tension: more numerous and more complex tasks, constrained finances, labour shortages, legal requirements, digital transformation, and rising pressures from climate, nature, and maintenance backlogs. The conventional reform response is to ask which tasks can be reduced, which processes can be automated, and which financial gains can be realised.

These questions may be necessary, but they often begin too late in the causal chain.

> Where is municipal capacity tied up in working relationships, handovers, systems, and chains of responsibility that do not create corresponding value?

The first yes requested of KL — Local Government Denmark is therefore not a yes to a new central model or a predetermined savings requirement. It is a yes to examining the question precisely through a small number of locally grounded and reversible pilots.

A legitimate pilot outcome may be to stop, redesign, or conclude that the work examined is necessary.

Four connected reform moves

Friction economics is a method for examining where capacity is tied up and whether some of that tie-up can be reduced responsibly.

G1/G2/G3 distinguishes capacity actually freed, documented avoided costs, and increased robustness or political room for manoeuvre. The categories prevent every change in time use from being treated as a cash saving.

Digital legal safeguards and the Orwell test place a legal and democratic control architecture around municipal AI and digitalisation. The core question is: Does the solution create control without corresponding insight, counter-power, and the possibility of repair?

Precise democratic sensing connects concrete observations from citizens, employees, and places with verification, responsibility, decision, feedback, and correction.

One local 90-day pilot format

Period	Primary task	Decision gate
Days 1–15	Mandate, scope, responsibility, and minimum baseline	Continue, redesign, or stop
Days 16–30	Precise baseline, field map, rights, and stop criteria	Intervention approved or stopped
Days 31–75	One reversible intervention and weekly correction loops	Continue, adjust, pause, or roll back
Days 76–90	Post-measurement, net correction, and G1/G2/G3 reading	Stop, retest, continue locally, or investigate scaling

Five possible decisions for KL

1. Recognise friction as a possible category of municipal capacity.
2. Establish a bounded 12-month cross-municipal learning track.
3. Invite five to ten municipalities into local 90-day pilots.
4. Adopt temporary AI, data, and supplier control safeguards.
5. Publish a shared learning and correction report.

In this report, the regenerative municipality is not a municipality that claims to have removed all friction. It is a municipality able to distinguish necessary complexity from reducible capacity tie-up, technical agency from human responsibility, and freed time from an actual financial gain.

> It is not faultless. It is repairable.

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3. Municipal Capacity Under Pressure

Municipal capacity cannot be read directly from a budget or from the number of staff hours. Two units with comparable resources may have radically different practical agency depending on their division of labour, decision authority, system support, professional stability, and ability to detect and correct errors.

Capacity therefore includes more than time. It also consists of overview, judgement, responsibility, trust, institutional memory, and the ability to understand what happens when a task moves between people, units, and digital systems.

A substantial share of municipal work takes place between the formal tasks. A citizen makes contact again because the next step is unclear. An employee records the same information in several systems. A case moves to another unit without responsibility and context travelling clearly with it. A meeting is repeated because those present lack decision authority. An error generates reprocessing, a complaint, and additional control.

Some of this is necessary work. Some protects the citizen and the municipal organisation against arbitrariness, professional error, and loss of legal safeguards. Record-keeping, reasons for decisions, the right to be heard, and professional quality control are not friction merely because they take time.

But some may be capacity tied up because the relationships within the work field do not function with sufficient clarity.

Four categories of work

Necessary core work carries out the municipal task: education, care, maintenance, guidance, administrative assessment, and professional coordination.

Necessary protective work safeguards legal rights, quality, safety, data protection, working conditions, and democratic accountability.

Potentially reducible friction is work that may be limited without weakening the core task or the level of protection.

Displaced friction is work or burden that appears to be removed in one place but is shifted to another employee, another unit, the citizen, relatives, a later budget year, or another physical place.

This distinction is crucial. A faster digital workflow is not a gain if the citizen subsequently has to do more work, if frontline staff have to correct more errors, or if the municipality loses the ability to understand and correct the supplier’s system.

Friction often arises at organisational interfaces: between citizen and system, professional practice and documentation requirements, separate units, political decision and operations, and municipal responsibility and a supplier's technical control.

This does not mean that coordination is superfluous. Coordination may be core work. The question is whether the relationship creates necessary coherence or repeatedly compensates for unclear responsibility.

A municipal capacity analysis must therefore ask not only what the organisation produced. It must also ask which human, institutional, and ecological capacity was preserved or degraded in the process.

An intervention may increase output in the short term while weakening professional judgement, working conditions, or the municipality's ability to change supplier. It may reduce a visible cost by postponing maintenance and thereby increase future damage.

> Friction is a question for investigation, not a judgement made in advance.

The investigation may conclude that the work should be preserved; that it should be organised more clearly; that some of it can be reduced; or that the proposed intervention is worse than the original problem.

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4. Evidence Base and Reform Proposition

The report's evidence base consists of different layers that must remain analytically distinct.

Official statistics, legislation, public-authority guidance, and external research can document municipal financial frameworks, existing uses of AI, legal obligations, and known problems of coordination and documentation. These sources do not document the report's integrated reform architecture.

The author's earlier municipal research on horizontal coordination and area-based interventions contributes an understanding of hidden coordination costs, cross-cutting responsibility, and the relationship between local knowledge and central decision-making. That research does not in itself validate the report's current AI or gain model. [K11]

The economic lineage also includes EVA's Pengene og livet [Money and Life], Jesper Jespersen's article on national accounts, economic models, and planetary survival, and Jesper Jespersen and Steen Brendstrup's Grøn økonomi [Green Economics]. They serve here as historical and conceptual lineage — not as evidence of municipal effects. [K12] [K13] [K14]

VIVE's analyses of local autonomy and free-institution experiments show, among other things, that rule simplification and reduced documentation are not frictionless processes: they may involve transaction costs and require a local mandate, leadership, and assessment of whether documentation actually supports professional practice. VIVE's more recent work on integrated interventions also shows substantial variation in

cross-cutting organisation and a need to distinguish forms of collaboration, professional goals, and financial goals. These findings support the need for bounded investigations, but they do not validate the report's integrated model. [K15a] [K15b] [K15c]

The Green Papers inquiry contributes work-field ecology, correction loops, shared green/yellow/red reading, stop rights, situated knowledge, and G1/G2/G3. These works are internal methodological precursors, not independent effects research.

Five falsifiable propositions

Proposition 1. Municipalities may have substantial capacity tied up at organisational interfaces that is not visible in conventional activity measures.

Proposition 2. Some of these tie-ups may be reducible through small and reversible interventions without weakening rights, quality, or working conditions.

Proposition 3. G1/G2/G3 may create stronger gain discipline than conventional business cases by separating freed capacity, avoided costs, and robustness.

Proposition 4. AI-assisted pattern recognition may help identify repeated friction if its use is purpose-bounded, responsibility-bearing, contestable, and reversible.

Proposition 5. KL may be able to standardise the architecture for learning and protection without standardising every local problem and solution.

The five propositions need not stand or fall together. A pilot track may strengthen one, disprove another, and show that a third requires substantial reformulation.

> It is a scientific and institutional strength of the report if practice is allowed to change the model.

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5. Friction Economics and Gain Discipline

Friction economics is a method for examining where municipal capacity is tied up, which part of that tie-up is necessary, whether some of it can be reduced responsibly, and how any resulting effect should be understood.

The method does not begin with a financial target. It begins with a concrete work field.

Friction is often relational. It arises when people, units, and systems cannot rely sufficiently on one another's information, responsibilities, assessments, and next actions. The result may be additional controls, repeated documentation, defensive workflows, and informal follow-up.

An intervention must therefore ask not only whether a step can be removed. It must examine which function the step actually performs.

Baseline and net correction

A gain claim requires a documented condition before the intervention. The baseline must describe the scale of the work field, relevant time or quality indicators, the people affected, and known secondary effects.

The baseline must be sufficient to read a change, but it must not become a new registration machine. Measurement must not develop into the friction it was meant to investigate.

A before-and-after difference does not necessarily document the effect of the intervention. Case volume, case complexity, season, staffing, legislative changes, other initiatives, and the pilot's own attention effect may influence the result.

Net correction does not mean that every pilot must establish statistically certain causality. It means that competing explanations must be visible and that the report must not claim more than the material can support.

> No gain without a baseline, and no account of effect without net correction.

G1: freed capacity

G1 is time, attention, calm, or judgement that actually becomes available through the intervention. It may consist of fewer avoidable repeat contacts, less rework, fewer transfers of responsibility, or fewer emergency operational responses.

G1 is not automatically cashable. Five minutes freed across many dispersed processes may improve the work without translating into fewer positions or a lower budget.

G2: avoided costs and harms

G2 is a documented or reasonably substantiated reduction in errors, complaints, reprocessing, operational interruptions, or later damage.

A possible event must not be booked as a certain saving. G2 must show which event may have been avoided, how often it normally occurs, and the degree of uncertainty.

G3: robustness and political room for manoeuvre

G3 is an increased ability to detect problems, locate responsibility, correct errors, make decisions, and preserve municipal freedom of action.

A clear stop mechanism, better insight into a supplier system, or a faster connection between local observation and responsible operations may be G3 outcomes.

G3 must not be assigned an arbitrary monetary value. Robustness may be politically valuable even when it cannot be monetised.

The categories cannot be added together mechanically

The same change may have several effects. Fewer errors may free time, reduce reprocessing costs, and strengthen the organisation's ability to detect systemic problems.

If the full value is counted in all three categories, double counting occurs. A local gain reading must therefore show what was observed, what was calculated, what was assumed, and where overlaps exist.

No fixed allocation percentages are used between G1, G2, and G3.

From time to actual realisation

Before freed time is treated as realisable capacity, the municipality must examine whether the change is stable, whether the time is concentrated or dispersed, whether the work has actually been removed, and whether new control or maintenance tasks have arisen.

Freed capacity should normally first be used to stabilise the field, restore quality, reduce strain, or strengthen the core task. Cash realisation requires a separate administrative and political decision.

> Freed time is not automatically a cash saving. Avoided risk is not the same as released payroll expenditure. Robustness must be documented without arbitrary monetisation.

National arithmetic sensitivity

For 2026, Danish municipalities budgeted service expenditure of DKK 336.7 billion and capital investment of DKK 22.7 billion. The preceding economic agreement set a service-spending framework of DKK 336.8 billion; the framework and the final budgeted service expenditure are two different official figures. [K1] [K2]

Illustrative share	Arithmetic gross value
0.05%	DKK 168.35 million
0.10%	DKK 336.70 million
0.25%	DKK 841.75 million
0.50%	DKK 1,683.50 million
1.00%	DKK 3,367.00 million

The table is not an estimate of actual municipal friction, an expected gain range, or a national savings potential. It shows only that a small arithmetic share of a large service economy can have a substantial order of magnitude.

> Field → baseline → intervention → post-measurement → net correction → G1/G2/G3 → decision.

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6. One Integrated 90-Day Pilot

Friction economics should not begin as a new permanent management system or an extensive data-collection exercise. It should begin with a bounded question in a concrete municipal work field:

> Can one specific item of friction be reduced through a small, reversible intervention without weakening legal safeguards, working conditions, trust, access to support, or the natural foundation affected?

The pilot should normally cover one work field, one primary item of friction, one institutional accountability-holder, a small number of baseline indicators, and one main intervention.

It may take place within a citizen pathway, a handover between two units, one type of letter, a meeting or approval process, a bounded digital function, or a specific physical place with repeated operational problems.

The field must be small enough for the intervention to be tested and rolled back within 90 days. It need not have the greatest expected financial potential. A smaller field with clear chains of responsibility is often a better first learning field than a large project with high political visibility.

Roles and responsibilities

A political or administrative mandate-holder approves the purpose and framework at a level appropriate to the pilot's content and risk.

A named institutional accountability-holder has genuine authority to ensure that rights, stop criteria, and decisions are respected. A project coordinator without decision authority is not sufficient.

A small professional field group brings together the necessary operational, professional, legal, and occupational-health knowledge. Where citizens or local conditions are affected, the relevant observational positions must be included. Citizen involvement never replaces the municipality's responsibility as a public authority.

A critical-friend function should be able to challenge assumptions, investigate displaced friction, and recommend a pause. It must not develop into a parallel management structure.

Days 1–15: mandate and minimum baseline

The first 15 days are used to formulate a short pilot mandate. It must describe the work field, the assumed item of friction, the people affected, the accountability-holder, the expected intervention, and possible harmful effects.

At the same time, the municipality must determine whether a usable minimum baseline exists. This may cover the number of incidents, repeat contacts, handovers, returned cases, time use, errors, or the citizen's understanding of the next step.

The purpose is not a full effects evaluation. It is to determine whether the field is sufficiently bounded and observable.

The pilot proceeds beyond day 15 only if there is a genuine accountability-holder, a practical possibility of stopping, a reasonable evidence base, and probable learning value proportionate to the burden imposed.

Days 16–30: field map and precise baseline

The work field is read through three questions. **Flow:** What must move through the process — information, responsibility, decision, support, or physical operations? **Friction:** Where may capacity be tied up without corresponding value? **Sensitivity:** Where could an error or delay have significant consequences for rights, care, working conditions, safety, personal data, or the natural foundation?

No more than three to five indicators should normally be selected. For each indicator, the municipality must describe its definition, data source, measurement period, known limitations, and competing explanations.

Before the intervention begins, concrete stop criteria must be established. They may include reduced access to support, a higher error rate, absence of legal compliance, unacceptable employee strain, discrimination, uncontrolled data use, or substantial displaced friction.

The intervention may begin only when the baseline, responsibility, rights, and rollback arrangements are sufficiently clear.

Days 31–75: intervention and correction

The pilot tests one main intervention. It may consist of a clearer letter structure, removal of one unnecessary handover, placement of decision responsibility, consolidation of duplicate registration, a changed operational routine, or a narrowly bounded AI-assisted function.

The intervention must be described precisely enough for others to see what actually changed. It must not gradually expand to new purposes, data types, or target groups without a new assessment.

The field group conducts a short weekly correction loop: What did we observe? What changed? Which errors or new burdens arose? Was the work displaced? Should the intervention continue, be adjusted, paused, or rolled back?

Observation, interpretation, and decision remain separate. A green status means that the intervention continues to function within the mandate. Amber means uncertainty or a need for adjustment. Red means signs of significant harm, breach of rights, uncontrolled function creep, or absence of a possibility of repair.

The colour applies to the field — not to the employee.

Days 76–90: post-measurement and decision

The post-measurement compares the indicators with the baseline and also describes qualitative observations, errors, displacements, costs, and data limitations.

The results are read as G1, G2, and G3, but the report must also show what was not a gain and what remains too uncertain to classify.

On day 90, one of four decisions is taken: stop and roll back; adjust and test again; continue cautiously at local level; or prepare a separate scaling investigation.

A positive local effect does not automatically document a lasting effect, national economic value, or scope for budget reduction.

The pilot’s primary output is a short learning note describing the field, baseline, intervention, effect, negative findings, G1/G2/G3, uncertainties, and the accountable decision.

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7. Digital Legal Safeguards and the Orwell Test

Municipal AI is no longer a future scenario. KL’s municipal AI map brings together municipalities’ own submissions about specific AI solutions and illustrates their use in municipal service delivery.

The map is based on municipalities' own submissions and descriptions. The map is therefore a knowledge bank — not a complete or independently validated account of all municipal uses of AI. [K3]

At the same time, 64 municipalities are participating in seven clusters concerned with scaling AI-assisted documentation. Municipal AI is therefore present both in local operations and in organised dissemination. [K4]

The report's contribution is therefore not another programme for more AI. It is an architecture of rights and correction around a development already under way.

The central question is not only whether the technology works. It is also which work disappears, which work is displaced, who is affected, who carries responsibility, and how errors can be detected, contested, and repaired.

Four status layers

Applicable law includes data-protection law, administrative law, sectoral legislation, occupational-health requirements, and the parts of the AI Act already in application.

Adopted requirements with later application dates include substantial parts of the high-risk regime.

Public-authority guidance includes material from the Danish Agency for Digital Government, the Danish Data Protection Agency, and the Parliamentary Ombudsman.

The municipal standards proposed by this report include the Orwell test, a public AI register, stop and rollback arrangements, and a strengthened standard of human responsibility.

The report's own standards must not be presented as applicable law. A municipality may, however, choose a higher level of protection than the legal minimum.

Timeline of the AI Act

The AI Act entered into force on 1 August 2024. Rules on AI literacy and the original prohibited practices have applied since 2 February 2025. Most of the rules, including the specific transparency requirements in Article 50, apply from 2 August 2026. The Digital Omnibus moved the principal high-risk requirements for Annex III systems to 2 December 2027 and for Annex I systems to 2 August 2028. [K5] [K6] [K7]

Providers of generative AI systems placed on the market before 2 August 2026 have until 2 December 2026 to comply with Article 50(2)'s machine-readable marking requirement.

The postponement does not mean that municipalities are free of obligations until then. Data-protection law, administrative law, sectoral legislation, and the municipality's general responsibility as a public authority continue to apply.

The applicable Danish supplementary act is Act No. 467 of 14 May 2025. The broader Bill L 111 was introduced on 18 February 2026 but lapsed and is not applicable law. [K8a] [K8b]

In the employment field, Executive Order No. 101 of 20 January 2026 provides municipalities with a sector-specific framework for processing personal data using AI or similar digital solutions. The executive order requires documentation and clear information to the data subject and does not permit decisions based solely on automated processing, including profiling. [K8c]

System design is also administrative design

A digital system can change which information becomes visible, which categories acquire administrative significance, and which exceptions can be processed.

When an error is built into a digital system, it can be reproduced systematically. Consistency can strengthen legal safeguards if the system is correct and can handle exceptions. It can also systematise an error.

The Danish Parliamentary Ombudsman emphasises that requirements of administrative law and the procedural rights of parties must be incorporated into public IT systems from the outset. Among other things, the system must be able to support the right to be heard, documentation, and relevant exception routes. [K10a] [K10b]

Digital legal safeguards therefore begin not at the user interface but with the system's purpose, categories, data sources, decision points, and chain of responsibility.

Data across the full life cycle

The AI Act does not replace the General Data Protection Regulation. The municipality must have a valid legal basis for processing, limit purpose and data volume, ensure data quality, and assess the need for a data protection impact assessment. [K9]

Development, training, and operation may constitute different purposes of processing. The municipality therefore cannot automatically assume that information collected for one municipal task may also be used for model development, pattern recognition, or control.

The impact assessment must not become a document prepared only after the technology, supplier, and data model have been locked in. It must be capable of changing or stopping the project.

Profiling and human control

Profiling, decision support, and a fully automated decision are different functions. Profiling evaluates or predicts personal characteristics. Decision support provides a recommendation or categorisation while a person makes the actual decision. A fully automated decision is made solely through automated processing and affects the person legally or in a similarly significant way.

Formal human approval does not necessarily make a system genuine decision support. If the employee does not understand the output, lacks time to make an independent assessment, or lacks authority to override it, the system may de facto direct the decision.

> The assessment must be individual, borne by human responsibility, reasoned, and genuinely contestable.

The Orwell test: control architecture, counter-power, and repair

The report's democratic stress test is:

> Does the solution create control without corresponding insight, counter-power, and the possibility of repair?

The name is not used to claim that municipal digitalisation is inherently totalitarian. The test names a specific risk: that the power to observe, connect, classify, and act grows faster than the affected person's ability to understand, contest, and stop its use.

Dimension	Control architecture is present when...	Repair architecture requires...
Visibility	The municipality can see the citizen while the system's own functioning remains unclear	The system's role, data, and accountability are visible
Purpose	Data drifts into new uses without a new assessment	New purposes require a new legal and democratic mandate
Classification	Scores and categories acquire significance without a means of correction	Categories can be explained and contested
Decision power	Human control consists only of formal approval	A competent person can genuinely override the system
Responsibility	Responsibility dissolves between supplier, system, and organisation	A named human and institutional accountability-holder
Insight	The affected person sees the outcome but not its basis or the next step	An intelligible explanation and a route for action
Contestability	An objection is recorded without any possibility of change	Genuine factual correction and human reassessment
Function creep	Pilot data becomes control or HR data	New uses require a new mandate
Displaced friction	A local gain creates burdens elsewhere	Secondary effects form part of the evaluation
Stop	The municipality cannot stop the system without the supplier	Local authority to stop and an alternative mode of operation
Rollback	A harmful change becomes permanent	The system and workflow can be restored to the earlier state
Repair	The error is registered but its consequences continue	Affected cases and data can be corrected
Democratic control	A significant use expands through routine operations	Expansion requires a legitimate new decision

The test is not a scoring model. A single serious red observation may be sufficient to require a pause.

> Is the municipality's growing capacity to observe, classify, and act accompanied by a corresponding capacity among citizens, employees, and democratic institutions to see, understand, contest, correct, and stop?

Stop and municipal freedom of action

A system is not genuinely controlled if the municipality cannot stop it.

Every significant use of AI should have a documented stop-and-rollback mechanism. It must describe who may trigger an investigation, who may decide on a pause, how ongoing cases are protected, which alternative workflow is activated, and which requirements apply before restart.

A municipality can purchase technology, but it cannot delegate away its responsibility. Supplier requirements must therefore cover the system’s purpose, data sources, version changes, audit access, known limitations, human override, data export, exit, and rollback.

The report also proposes a public municipal register of significant AI uses, stating purpose, accountability-holder, professional areas affected, data use, human control, complaint route, and stop mechanism.

The register does not replace individual information or reasons for decisions. It makes the municipal AI infrastructure politically and democratically visible.

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8. Precise Democratic Sensing

Municipalities already receive large volumes of information from citizens, employees, complaints, operational systems, inspections, and political enquiries.

The problem is often not a lack of input, but that observations lose their context, are not connected with other observational positions, or do not reach a function that has both responsibility and authority to act.

> Precise democratic sensing means that concrete observations from citizens, employees, and places can be recorded, investigated, connected to responsibility, and carried through to a visible municipal action or reasoned decision.

The method does not replace representative democracy, professional assessment, administrative law, consultation, or rights of complaint. It makes the municipality’s basis for sensing more precise and the trail of responsibility more visible.

A general judgement such as “the municipality’s digital service is poor” is difficult to investigate. A concrete description of a letter, a date, a missing next step, and three different answers by telephone can be verified.

Precision does not make the observation infallible. It makes it correctable.

Observation is not a vote

The number of observations does not automatically determine what the municipality should do. Ten similar observations may indicate a pattern. A single observation may be decisive if it reveals a serious breach of rights or a safety risk.

Conversely, many similar enquiries may arise from a shared misunderstanding or an organised campaign. Observations must therefore be read according to precision, consequence, position, documentation, and uncertainty — not volume alone.

The responsibility trail

Stage	Question
Observation	What has concretely been seen, experienced, or measured?
Context	Where and under which conditions did it arise?
Consequence	Who or what was affected?
Verification	What supports or contradicts the observation?

Stage	Question
Responsibility	Who has authority to investigate or act?
Decision	What should the municipality do, and at which level?
Feedback	What happened to the observation, and why?
Correction	Did the response work, or must the decision change?
Learning	Is this an isolated event or a structural pattern?

Observation, interpretation, and decision must remain separate.

“The citizen could not identify the next step” is an observation. “The structure of the letter creates repeat contacts” is an interpretation. “The municipality will test a new letter structure” is a decision.

The distinction makes it possible to correct assumptions and avoid treating an AI-generated pattern as fact.

Multiple observational positions

No single position sees the whole field. The citizen sees the consequence of an unclear process. The employee sees the manual detours. The manager sees patterns across units. The lawyer sees legal basis and rights. The politician sees priorities and conflicts of value.

Multiple positions do not mean that everyone has the same decision authority. They mean that the decision should not be made as though only one position existed.

Participation can improve a decision. It cannot conceal the decision-maker.

AI as provisional pattern recognition

AI may be able to help identify recurring patterns in enquiries, complaints, error reports, and operational observations.

But the output is not in itself a democratic observation. The system may miss rare events, conflate different problems, or amplify bias in the material.

AI-assisted pattern recognition must therefore be used only as provisional sorting and investigative support. The pattern must be traceable back to the material, readable by people with knowledge of the field, and open to rejection or correction.

> AI can help formulate a question. AI cannot determine what an observation means democratically or legally.

The citizen’s next step

One simple measure of municipal precision is whether the citizen can see the next step.

After contact with a municipality, the citizen should as far as possible be able to understand what the municipality has decided, why, what will now happen, who carries responsibility, and how an error can be corrected.

Repeated enquiries may therefore signal more than unclear language. They may reveal unclear responsibility, conflicting requirements, or a digital system that does not fit the citizen’s situation.

The method must not develop into sentiment analysis, hidden profiling, or a channel through which the municipality collects experience without any duty to respond.

> Democratic quality does not lie in the municipality seeing everything. It lies in the municipality seeing what is relevant without depriving the citizen or employee of the right to look back.

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9. Ecological Capacity as a Municipal Pilot Field

The preceding chapters have addressed municipal capacity in organisations, digital systems, and democratic trails of responsibility. The same logic of correction applies to the municipality's physical places.

A place may send early signals concerning heat, water, soil, vegetation, and maintenance without those signals being connected to responsible municipal action. The municipality may consequently compensate repeatedly for physical or ecological degradation in the same way that employees compensate for an unclear organisational workflow.

Ecological capacity is therefore introduced not as a parallel nature programme, but as a necessary extension of the report's fundamental question:

> Where is future municipal capacity tied up or degraded because a local signal is not seen, carried by responsibility, and corrected in time?

Municipal capacity also depends on soil, water, vegetation, buildings, roads, and local temperature conditions. When these systems lose function, municipal secondary effects arise: emergency responses, operational disruption, damage, maintenance needs, complaints, and pressures on health or working conditions.

> Ecological capacity is the ability of a municipality and a place to absorb pressure, maintain conditions for life, and restore function without repeated emergency intervention.

The concept must not be used to turn nature into a municipal service provider or reduce all ecological values to money. It should make the connection between local natural conditions, municipal operations, and future capacity relevant to decision-making.

The place as the unit

A pilot field may be a schoolyard, a care home, a stretch of road, a green space, a local low point, or a building with recurring heat problems.

The scope must show which place is being examined, who is affected, which units carry responsibility, and which effects can be observed within 90 days.

The place is not a closed system. A local intervention may displace water, heat, or a maintenance burden to another place. Displaced friction must therefore also be investigated physically and ecologically.

From signal to responsibility

Local citizens and employees can often see where water collects, which trees are weakening, or when a room becomes unusably hot before the problem appears in central systems.

The observations are important but not self-sufficient. They must meet technical and ecological expertise, maps, historical incidents, weather measurements, and other observational positions.

The ecological responsibility trail follows the same structure as democratic sensing: observation, place, consequence, verification, responsibility, action, feedback, and post-measurement.

A local observation system has little value if the municipality can receive photographs and markers but cannot connect them to an accountable operational function.

A bounded stormwater example

A pilot field may be a place with repeated flooding. The hypothesis may be that the municipality repeatedly uses capacity on clean-up, enquiries, and internal clarification without connecting observations and responsibility to a smaller preventive intervention.

The baseline may include incidents, enquiries, response time, operational hours, and known water pathways. The intervention may consist of cleaning, changed inspection, a small terrain adjustment, or clearer allocation of responsibility.

The pilot must also examine whether water is displaced to a more vulnerable place and whether the local solution conceals a need for a larger investment.

G1 may consist of fewer emergency operational hours. G2 may consist of fewer documented instances of damage. G3 may consist of better local knowledge and a faster municipal response.

The absence of flooding over a short period does not document a lasting effect.

A heat field

Another pilot field may be recurring heat stress at a school or care home. At present, the institution may compensate through relocated activities, improvised shading, and repeated enquiries without a shared picture of the place’s temperature, patterns of use, and responsibilities.

A small intervention may consist of temporary shade, changed use of areas, or a precise operational and escalation route.

Health, vulnerable users, working conditions, water consumption, and the risk of concealing a larger building problem must form part of the assessment.

A 90-day process may improve observation and response. It cannot document long-term climate effects or full ecological regeneration.

Ecological robustness must not be assigned an arbitrary monetary value. But a better basis for a long-term investment may be a significant G3 outcome.

10. KL’s Role and Five Possible Decisions

KL already has a central role in municipal digital strategy, knowledge-sharing, and scaling collaborations. The report therefore does not propose a parallel AI or digitalisation structure.

It proposes that KL strengthen the shared capacity to investigate friction, protect rights, read gains, and correct course.

KL can establish a shared method, learning format, and contractual standards, but it cannot take over the legal and democratic responsibility of the municipal council, the administration, or the specific public authority.

A local municipality can investigate a concrete field, but acting alone it has greater difficulty determining whether the same mechanism exists elsewhere, whether results are comparable, or whether a supplier’s claims hold across municipalities.

KL’s particular role is to make different experiences jointly readable without making them identical.

Six cross-municipal functions

Function	Cross-municipal task	What remains local?
Shared method	Minimum definitions of friction, baseline, and G1/G2/G3	Field, indicators, and intervention
Rights	Minimum standards for insight, responsibility, contestability, and stop	Concrete legal assessment
Learning	Shared format for positive and negative findings	Local interpretation
Supplier requirements	Audit, version changes, exit, and rollback	Contract and local use
Gain discipline	Distinction between capacity, cost, and robustness	Local budget decision
Transparency	Minimum standards for public pilot and AI information	Protected data and local communication

Comparison across municipalities must not become a ranking by the largest calculated gain or the greatest number of automated workflows. That would create incentives for optimistic business cases, hidden costs, and under-reporting of negative results.

The shared reading should instead examine whether the hypothesis was precise, whether the intervention was reversible, whether negative findings became visible, and whether work was displaced.

A pilot that shows precisely why an expected gain could not be realised may have greater shared value than a pilot with a large but uncertain gain figure.

Five possible decisions

1. RECOGNISE FRICTION AS A POSSIBLE CAPACITY CATEGORY

The first yes is a yes to investigating whether municipal capacity may be tied up in working relationships that are not visible in conventional measures. That recognition must be accompanied by a protection: friction must not become a general label for professional complexity, legal safeguards, or employees who are not working fast enough.

2. ESTABLISH A 12-MONTH LEARNING TRACK

The track should have a public, time-limited mandate and be organised as a small shared function, not a new extensive secretariat. Existing KL environments concerned with digitalisation, law, and professional sectors should be involved so that the track does not become a parallel structure.

3. INVITE FIVE TO TEN PILOT MUNICIPALITIES

The municipalities should represent different sizes, professional sectors, and organisational capacities. Pilot selection should include both digital and non-digital interventions and at least one ecological operational field. The purpose is not statistical representativeness, but to investigate which conditions affect the method's usefulness.

4. ADOPT TEMPORARY CONTROL SAFEGUARDS

Before the pilots begin, there should be a shared minimum standard for claim status, data minimisation, human responsibility, employee protection, the Orwell test, stop, rollback, and supplier requirements. The safeguards are themselves among the elements to be tested. They may subsequently be retained, strengthened, simplified, or rejected.

5. PUBLISH A LEARNING AND CORRECTION REPORT

The final report must not be a collection of success stories. For each field, it must show the original proposition, baseline, intervention, observed effect, implementation costs, displaced friction, negative findings, and methodological limitations.

The cross-cutting analysis should ask in particular which kinds of friction could actually be investigated, where measurement itself became a burden, and which parts of the reform proposition were disproved.

After 12 months, an explicit decision must be taken to close the track, investigate selected elements further, revise the method, or prepare a more narrowly bounded cross-municipal practice.

None of these outcomes should be automatic.

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11. Concluding Proposition

Municipalities will continue to make priorities under constrained financial, human, and ecological conditions. This report does not remove that reality.

It does not promise that municipal funding pressures can be resolved through friction reduction. It does not document a specific national potential and does not claim that AI in itself frees capacity.

> Municipalities may have substantial capacity tied up in organisational friction that conventional budget and activity measures do not make sufficiently visible. This possibility should be investigated through small, reversible pilots with rights safeguards before conclusions are drawn about gain, scaling, or permanent governance.

Friction economics first changes the direction of reading. It asks not only what a task costs, how quickly it is completed, or which technology can automate it.

It also asks where capacity is tied up, which work is necessary protective work, who bears hidden costs, and whether a change in time use actually becomes usable capacity.

G1/G2/G3 prevents all gains from being treated as cash savings. The Orwell test prevents technical efficiency from becoming the sole criterion of digital development. Precise democratic sensing connects local observations with verification, responsibility, and correction. The ecological dimension reminds us that municipal welfare also depends on water, soil, buildings, maintenance, and the capacity of places to absorb pressure.

The 90-day pilot makes the proposition practical and falsifiable. The KL track makes local learning shared without making every municipality the same.

> No gain without a baseline. No account of effect without net correction. No technological scaling without responsibility, contestability, and stop. No budget harvesting before the field has been stabilised.

The regenerative municipality is not the municipality that claims to have removed all friction. It is the municipality able to distinguish necessary complexity from reducible capacity tie-up; freed time from real savings; observation from conclusion; and technical agency from human responsibility.

It is not faultless. It is repairable.

It can see when a solution displaces work instead of removing it. It can stop when an experiment causes harm. It can reinvest capacity before harvesting it. It can make an error visible without first making a person guilty. And it can allow actual municipal practice to correct the propositions of reform.

The report’s first invitation to KL is therefore modest:

> Say yes to investigating the question precisely.

Not through a central total model. Not through a national savings promise. But through shared safeguards, a small number of local pilots, public learning, and a genuine possibility of stopping, correcting, or proceeding.

That is the first yes.

Appendix A. Core Concepts

Capacity is the municipality’s practical ability to perform core tasks, exercise professional judgement, coordinate responsibility, correct errors, and maintain the human, institutional, and ecological conditions for future public-service delivery.

Friction is capacity that is tied up, displaced, or degraded without correspondingly improving the citizen’s situation, the core task, legal safeguards, working conditions, or organisational robustness.

Necessary protective work is work that safeguards legal rights, professional quality, safety, data protection, working conditions, or democratic accountability.

Potentially reducible friction is a capacity tie-up that may be reducible without weakening the core task or the necessary level of protection.

Displaced friction is work or burden reduced in one place but shifted to other people, units, authorities, points in time, or physical places.

Baseline is the documented condition before the intervention.

Net correction is the assessment of which factors other than the intervention may explain an observed change.

G1 is capacity actually freed.

G2 is documented or reasonably substantiated avoided costs and harms.

G3 is increased robustness, accountability, or political capacity to act.

Cash realisation is the part of a stable, documented, and disposable change that can be translated into a budgetary decision without creating hidden secondary effects.

Control safeguard is a condition, boundary, or protection established in advance to prevent breaches of rights, harm, function creep, or uncontrolled scaling.

Stop-and-rollback mechanism is the documented ability to pause a use, activate an alternative workflow, and restore practice to a known and defensible state.

Repair architecture is an institutional and technical architecture in which errors can be detected, contested, investigated, corrected, and, as far as possible, remedied.

Precise democratic sensing is a practice in which concrete observations can be verified, connected with responsibility, and carried through to a visible action or reasoned decision.

Ecological capacity is the ability of a place and a municipality to absorb pressure, maintain function, and recover without repeated emergency intervention.

Appendix B. Calculation Discipline

Stage	Content
Baseline	Scale before the intervention
Post-measurement	Scale after the intervention
Observed difference	Raw before-and-after change
Net correction	Correction for concurrent factors
Gross value	Arithmetic conversion
Secondary effects	New burdens and displacement
Implementation cost	Development, operation, and evaluation
G1/G2/G3	Classification of the effect
Reinvestment	Capacity that should be returned
Realisation	Separate decision on disposable capacity

A time-based module may be calculated as: number of relevant events × net reduction in minutes ÷ 60 × documented local hourly cost.

The calculation must show volume, unit, period, and local hourly value. Implementation costs and displaced friction are deducted.

Overlapping events must not be double-counted. An unclear letter may, for example, lead to repeat contact, handover, reprocessing, and complaint. The entire sequence must not be valued four times.

Financial figures are labelled as observed, calculated, estimated, projected, or illustrative.

A public learning note must state that a locally calculated result does not automatically document a lasting effect, an effect in other municipalities, or scope for budget reduction.

Appendix C. Legal and Democratic Matrix

Status markers: **R** = applicable law; **F** = adopted requirement with a later application date; **V** = public-authority guidance; **S** = standard proposed by this report.

Theme	Status	Minimum before pilot or operation	Red signal
Problem and necessity	S/V	Describe the problem and simpler alternatives	The technology is selected before the problem
Function and role	R/F/V	Describe the system’s actual function and the municipality’s role	The supplier’s marketing replaces the municipality’s own assessment
Purpose and legal basis	R	Establish purpose, sectoral law, and legal basis for processing	“Efficiency” is stated as the sole purpose
Data	R	Minimise data and assess quality and special categories	Data is collected just in case
Life cycle	R/V	Assess development, training, operation, and changes separately	The legal basis for operation is assumed to cover training
DPIA	R/V	Assess the need before design and procurement are locked	The assessment is completed after deployment
Administrative law	R/V	Ensure adequate case information, the right to be heard, reasons, and exceptions	The system cannot handle lawful exceptions
Profiling	R/F	Identify evaluation of personal characteristics	Profiling takes place without a clear mandate
Automated decisions	R	Assess Article 22 and genuine human involvement	The person cannot override the output
Human responsibility	R/F/S	Ensure competence, time, information, and authority	Control consists of a click
Transparency	R/F/S	Explain the system’s role, responsibility, and next step	The citizen cannot see the significance of AI
Contestability	R/S	Create a route to factual correction and reassessment	Errors can be reported but not changed
Employees	R/S	Prohibit hidden performance control and secondary HR use	Friction data is used for ranking
Supplier	R/F/S	Require audit, version history, exit, and stop	The supplier can change the function unilaterally
Logging	R/F/S	Make relevant inputs, outputs, and decisions traceable	The municipality cannot reconstruct an error
Stop	S	Designate stop authority and alternative operation	The municipality cannot stop the system
Public visibility	S	Register significant AI uses	The use is politically invisible
Scaling	R/F/S	Carry out a new assessment for a new purpose or target group	Pilot approval is treated as scaling approval

The legal matrix asks which requirements apply. The Orwell test also asks whether municipal control capacity is accompanied by insight, counter-power, and repair.

Appendix D. Source Basis

The sources are divided by function. Primary law and official sources support legal, financial, and institutional facts. External research supports bounded empirical findings. The author's earlier research and the Green Papers provide lineage and method, but not independent validation of the report's integrated proposition.

K-markers are used for concrete factual or research-based claims in the main text. Green Papers and companion reports are accessible through their direct links. The listed versions come from the report's development and later reference updates; the list is not a complete catalogue of the library's current editions.

Official financial and municipal sources

K1. Danish Ministry of Finance. (2026). Kommunernes og regionernes budgetter for 2026 er opgjort [Municipal and regional budgets for 2026 have been compiled]. Final municipal budget figures: DKK 336.7 billion in service expenditure and DKK 22.7 billion in capital investment.

K2. Government of Denmark and KL — Local Government Denmark. (2025). Aftale om kommunernes økonomi for 2026 [Agreement on municipal finances for 2026]. Agreed municipal service-spending framework of DKK 336.8 billion.

K3. KL Knowledge Centre. Kommunernes AI-landkort [Municipal AI map]. Municipalities' own reported AI projects and status; accessed on 16 October 2026.

K4. KL. (2026). 7 klynger, 64 kommuner – nu skal AI-dokumentation skaleres [Seven clusters, 64 municipalities — AI documentation is now to be scaled].

Law and public-authority guidance

K5. Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence.

K6. Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 simplifying the implementation of harmonised rules on artificial intelligence — the Digital Omnibus on AI.

K7. Danish Agency for Digital Government. Reglerne i AI-forordningen [Rules in the AI Act] and AI Omnibus. Used for the current Danish overview of application dates.

K8a. Act No. 467 of 14 May 2025 on supplementary provisions to the Regulation on artificial intelligence.

K8b. Danish Parliament. L 111 — Bill on supplementary provisions to the Regulation on artificial intelligence. Introduced on 18 February 2026; status: lapsed.

K8c. Executive Order No. 101 of 20 January 2026 on municipalities' processing of personal data in connection with the use of artificial intelligence or similar digital solutions in employment services.

K9. Danish Data Protection Agency. (2023). Offentlige myndigheders brug af kunstig intelligens — Inden I går i gang [Public authorities' use of artificial intelligence — Before you begin].

K10a. Danish Parliamentary Ombudsman. Partsrettigheder og offentlige it-systemer [Procedural rights of parties and public IT systems].

K10b. Danish Parliamentary Ombudsman. Generelle forvaltningsretlige krav til offentlige it-systemer [General administrative-law requirements for public IT systems].

Research and methodological lineage

K11. Engberg, Lars A. (2008). Den horisontale søjle: Et strategisk udviklingsperspektiv for koordinering af områdeindsatser i Københavns Kommune [The Horizontal Pillar: A strategic development perspective on coordinating area-based interventions in the City of Copenhagen]. SBi 2008:16. Hørsholm: SBi Forlag.

K12. Andelsselskabet EVA. (1990). Pengene og livet: EVA's årsrapport '90 [Money and Life: EVA's annual report 1990]. Frederiksberg: Samfundslitteratur.

K13. Jespersen, Jesper. (1990). "Om nationalregnskab, økonomiske modeller og klodens overlevelse" [On national accounts, economic models, and planetary survival]. In Pengene og livet: EVA's årsrapport '90.

K14. Jespersen, Jesper & Steen Brendstrup. (1994). Grøn økonomi: En introduktion til miljø-, ressource- og samfundsøkonomi [Green Economics: An introduction to environmental, resource, and social economics]. Copenhagen: DJØF Publishing.

K15a. Hjelmar, Ulf & Lars Bo Pedersen. (2024). Fem spor i lokal frisættelse — 12 års danske erfaringer [Five tracks of local autonomy — 12 years of Danish experience]. Copenhagen: VIVE.

K15b. Bjørnholt, Bente & Christina Holm-Petersen. (2024). Erfaringer med friinstitutioner [Experience with free institutions]. Copenhagen: VIVE.

K15c. Dalsgaard, Camilla T., Kasper Lemvigh & Rasmus Højbjerg Jacobsen. (2026). Helhedsorienterede indsatser — Begrebsafklaring og praksiseksempler [Integrated interventions — Conceptual clarification and examples from practice]. Copenhagen: VIVE.

Green Papers and companion reports

K16. Engberg, Lars A. (2026). Municipal Work as Nature. Report 01, v1.0, February 2026.

K17. Engberg, Lars A. (2026). Eve & Adam, and the Penguins. v1.1, August 2026.

K18. Engberg, Lars A. (2026). The Correction Loop: Preserving Human Accountability in AI-Assisted Work. Report 02, v1.0, July 2026.

K19. Engberg, Lars A. (2026). Penguin Dashboard: Legibility as Governance. Report 04, version 0.4, July 2026.

K20. Engberg, Lars A. (2026). Knowing From the Ground. v1.0, June 2026.

K21. Engberg, Lars A. (2026). Regenerative Reciprocity. Report 06, v0.8, September 2026.

K22. Engberg, Lars A. (2026). Moral Biology. Green Paper 01, v0.1, January 2026.

Appendix E1. Pilot Mandate

Municipality	
Pilot title	
Work field or place	
Primary item of friction	
Pilot period	
Institutional accountability-holder	
Professional pilot lead	
Mandate-holder	
Participating units	
Citizens, employees, or places affected	
Critical-friend function	

Pilot question

Can _____ be reduced through _____ without weakening _____?

Field map

Flow: What must move through the field?

Friction: Where is capacity assumed to be tied up?

Sensitivity: Where could errors have significant consequences?

Baseline

Indicator	Definition	Data source	Period	Limitation
1				
2				
3				
4				
5				

Rights and stop

The pilot documents a lawful and precise purpose, data minimisation, relevant legal assessment, a genuine chain of human responsibility, insight and correction, employee protection, stop authority, alternative operation, technical and organisational rollback, and a completed Orwell test.

Appendix E2. Weekly Correction Log

Week and date	
Observation	
Interpretation	
Decision	
Displaced friction	
Status	Green / yellow / red

Next accountable action

A green status means continued operation within the mandate. Amber means uncertainty or a need for adjustment. Red means pause, investigation, or rollback. The colour applies to the field — not to the employee.

Appendix E3. Final Learning Note

The learning note should briefly describe the work field, friction hypothesis, baseline, intervention, post-measurement, negative findings, displaced friction, G1/G2/G3, uncertainties, and the accountable decision.

Gain reading

Effect	Observed / calculated / estimated	G1	G2	G3	Uncertainty

Implementation costs:

Capacity that should be reinvested:

Possible displaced friction:

What the material cannot document:

Decision on day 90

— Stop and roll back.
— Adjust and conduct a new bounded test.
— Continue cautiously within the same local field.
— Prepare a separate scaling investigation.

Decision taken by:

Reason:

Date:

> The pilot material documents a bounded local process. The results do not automatically demonstrate a lasting effect, an effect in other municipalities, a national gain potential, or scope for budget reduction.

Related Documents

— Municipal Work as Nature
— Eve & Adam, and the Penguins
— The Correction Loop
— Penguin Dashboard
— Knowing From the Ground
— Regenerative Reciprocity

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